

List and Tuple – lab work

1. E-Commerce Order Analysis

Problem Statement

An online store records orders as:

```
orders = [  
    ("Laptop", 55000),  
    ("Mouse", 800),  
    ("Keyboard", 1500),  
    ("Monitor", 12000),  
    ("Pen Drive", 600)  
]
```

Write a program to:

- Display all products costing more than ₹1000.
 - Find the most expensive product.
 - Calculate the total order value.
 - Count products costing below ₹1000.
-

2. Cricket Tournament Scorecard

Problem Statement

A batsman's scores in different matches are stored in a list.

```
scores = [45, 78, 12, 100, 67, 8, 90, 55]
```

Write a program to:

- Count half-centuries and centuries.
 - Find the highest score.
 - Display all scores below 20.
 - Calculate the average score.
-

3. Smart Parking System

Problem Statement

Parking slots are represented as:

```
slots = [1, 0, 1, 1, 0, 0, 1, 0]
```

Where:

- 1 = Occupied
- 0 = Available

Write a program to:

- Count occupied and available slots.

- Find the first available slot.
 - Display all available slot numbers.
 - Check whether parking occupancy exceeds 75%.
-

4. Employee Salary Processing

Problem Statement

Employee data is stored as tuples:

```
employees = [  
    ("Rahul", 35000),  
    ("Priya", 55000),  
    ("Amit", 42000),  
    ("Neha", 65000)  
]
```

Write a program to:

- Display employees earning above ₹50,000.
 - Find the highest-paid employee.
 - Calculate total salary expenditure.
 - Count employees earning below ₹40,000.
-

5. Library Book Search

Problem Statement

Books available in a library:

```
books = [  
    ("Python Basics", 5),  
    ("Data Science", 0),  
    ("Java Programming", 3),  
    ("Machine Learning", 0)  
]
```

Write a program to:

- Display unavailable books.
 - Find all books with more than 2 copies.
 - Count available books.
 - Stop searching once a requested book is found.
-

6. Student Attendance Tracker

Problem Statement

Attendance for 15 days is recorded as:

```
attendance = ['P', 'P', 'A', 'P', 'A', 'P', 'P', 'P', 'A', 'P', 'P', 'A', 'P', 'P', 'P']
```

Write a program to:

- Count present and absent days.
 - Calculate attendance percentage.
 - Determine eligibility (minimum 75% attendance).
 - Display positions where the student was absent.
-

7. Online Quiz Evaluation

Problem Statement

Correct answers:

correct = ['A', 'C', 'B', 'D', 'A']

Student answers:

student = ['A', 'B', 'B', 'D', 'C']

Write a program to:

- Calculate score.
 - Display incorrectly answered question numbers.
 - Count correct and wrong answers.
 - Determine pass/fail (minimum 60%).
-

8. Bus Route Monitoring

Problem Statement

Passenger count at each stop:

passengers = [12, 18, 25, 30, 28, 15, 8]

Write a program to:

- Find the busiest stop.
 - Display stops with fewer than 10 passengers.
 - Calculate average passengers.
 - Determine whether any stop exceeded 25 passengers.
-

9. Warehouse Product Inspection

Problem Statement

Product IDs and quality status:

```
products = [  
    (101, "Pass"),  
    (102, "Fail"),  
    (103, "Pass"),  
    (104, "Fail"),
```

```
(105, "Pass")  
]
```

Write a program to:

- Display failed product IDs.
 - Count passed and failed products.
 - Calculate pass percentage.
 - Stop checking if 3 failures are found.
-

10. Train Reservation Waiting List

Problem Statement

Passenger records:

```
passengers = [  
    ("Anuj", "Confirmed"),  
    ("Rahul", "Waiting"),  
    ("Priya", "Confirmed"),  
    ("Amit", "Waiting"),  
    ("Neha", "Confirmed")  
]
```

Write a program to:

- Display all waiting-list passengers.
- Count confirmed and waiting passengers.
- Find whether a specific passenger has a confirmed ticket.
- Create separate lists for confirmed and waiting passengers.